

Ferric Acrisols

Ferric Acrisols are strongly weathered tropical soils characterized by a clay-rich subsurface horizon (argic horizon) and a low base saturation (<50%) within this zone.

The term “*Ferric*” refers to the **high iron content** and distinct reddish to yellowish-brown coloration due to iron oxide accumulation.

These soils are typically acidic, leached, and nutrient-poor, formed under warm, humid climates on old, stable landscapes such as uplands and plateaus. They commonly develop on weathered parent materials like granites, sandstones, or old alluvial deposits, with kaolinitic clay dominating their mineral composition.

Drainage:

- **Wet season:** Well-drained; surface runoff is moderate to high on sloping terrain.
- **Dry season:** Retains little moisture due to low organic matter and coarse texture in upper horizons.
- **Landscape:** Commonly found on **gently undulating to hilly terrain** in humid and sub-humid tropical zones; prone to erosion when vegetation cover is lost.

Depth:

- Generally **deep (100–200 cm)**, but effective rooting depth may be restricted by compact argic horizons or accumulation of iron concretions.
- Rooting is often shallow in eroded or strongly leached profiles.

Workability:

- **When wet:** Plastic and sticky; easily puddled by rainfall or tillage; risk of compaction if worked when saturated.
- **When dry:** Hard and firm; tillage can be difficult without prior moistening.
- Despite these constraints, Ferric Acrisols have **moderate physical stability** when well-managed and protected from erosion.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–15 cmol(+)/kg), dominated by kaolinitic clay and sesquioxides.
- **Base saturation:** Low (<50%), reflecting advanced weathering and leaching.
- **Nutrients:** Nitrogen, phosphorus, and potassium are **deficient**; phosphorus fixation is **very strong** due to Fe and Al oxides.
- **Organic matter:** Low to moderate (1–2%); declines rapidly under continuous cultivation.
- **Micronutrients:** Possible deficiencies in boron and zinc; iron and manganese usually abundant but may reach toxic levels in poorly drained sites.

pH:

- **Strongly to moderately acidic (4.5–5.5);** acidity increases with leaching intensity and low base content.

Management:

- Incorporate **organic matter** (compost, green manure, crop residues) to improve structure, water retention, and nutrient status.
- Apply **lime** to correct acidity and improve base saturation.
- Apply **phosphorus fertilizers** in small, frequent doses or use rock phosphate to counter strong P fixation.
- Use **cover crops** and **mulching** to minimize erosion and maintain soil organic carbon.
- On slopes, practice **terracing or contour planting** to control runoff.
- Suitable for **cassava, oil palm, tea, rubber, maize, and upland rice**, especially when managed with integrated fertility improvement strategies.
- Agroforestry systems combining trees and crops can help sustain long-term fertility and reduce erosion.